

ECE 635 Homework #6 Answers

1. Two components in the electric field calculation will cancel out thereby removing the radial component of the electric field.

2. $\hat{x} = \sin\theta \cos\phi \hat{r} + \cos\theta \cos\phi \hat{\theta} - \sin\phi \hat{\phi} \leftarrow$

(a) from $x = r \cos\theta \cos\phi \rightarrow \nabla x = \nabla(r \cos\theta \cos\phi)$

similarly for y and z

(b) $\hat{\phi} = (\hat{\phi} \cdot \hat{x})\hat{x} + (\hat{\phi} \cdot \hat{y})\hat{y} + (\hat{\phi} \cdot \hat{z})\hat{z}$

put $\hat{x}$, $\hat{y}$ and $\hat{z}$ from part (a) into this equation

$$\hat{\phi} = -\sin\phi \hat{x} + \cos\phi \hat{y}$$

$\rightarrow 3. (a) E_{\theta} = \frac{j k \eta I_0 l}{8\pi} \frac{e^{-jkr}}{r}$

Hertzian from Notes

(b) $\bar{P}_{ave} = \frac{\eta}{2} \left(\frac{I_0 l k}{8\pi} \right)^2 \frac{\sin^2\theta}{r^2} \hat{r}$

(c) $P_{rad} = 10\pi^2 I_0^2 \left(\frac{l}{\lambda} \right)^2$

(d) $R_{rad} = 20\pi^2 \left(\frac{l}{\lambda} \right)^2$

(e) $D_0 = 1.5$

(f) no loss $\Rightarrow G_0 = D_0 = 1.5$

4. $\bar{A} = \frac{j I_0 (2\pi a)}{4\pi} \frac{e^{-jkr}}{r} J_1(ka \sin\theta) \hat{\phi}$

For a small loop $ka \sin\theta \ll 1 \Rightarrow$ use the small argument approximation given for the Bessel function

$$J_1(x) \approx \frac{x}{2}$$

$$\therefore A = \frac{I_0 (\pi a^2)}{4\pi} \left(\frac{j k e^{-jkr}}{r} \right) \sin\theta \hat{\phi}$$